Pulmonary embolism: High-and intermediate-risk

Objective:

To provide a systematic and evidence-based approach in the identification and treatment of patients with high-risk and intermediate-risk acute pulmonary embolism (PE).

Background:

Acute PE varies in presentation and prognosis. As the severity of PE presentation increases, it carries increasing complexity, morbidity, and mortality. The first step when assessing an acute PE is **risk stratification,** which then determines the degree to which escalated and advanced therapies are needed to mitigate PE-related complications and death. The European Cardiology Society (ESC) has published guidelines helping to subdivide acute PE into low risk, intermediate-low risk, intermediate-high risk and high risk based on hemodynamics, Pulmonary Embolism Severity Index (PESI) or simplified PESI (sPESI) score and signs of right ventricular (RV) strain (**Table 1**).

Table 1: ECS Classification of PE severity and the risk of early (in-hospital or 30-day) death

* Persistent systolic blood pressure (SBP) <90 mmHg or drop >40 mm Hg from baseline; need for vasoactive or inotropic support to maintain SBP >90 mmHg

Low risk PE comprises 40-60% of acute PE diagnoses and carries a 30-day mortality of 1%; intermediate-risk PE comprises 35-55% of acute PE diagnoses and carries a 90-day mortality of 3-15%; high-risk PE comprises 5% of acute PE diagnoses and carries an in-hospital mortality of 30%. When PE is associated with c

acute mortality is 80%. Because of this wide spectrum of disease presentation and prognosis, an essential step in determining subsequent management for acute PE. It is important to note are not static. Hemodynamic deterioration can occur after initial assessment, ie, progressing f into high-risk, or from high-risk into cardiac arrest. Similarly, patients with intermediate-risk PE can also stabilize with early treatments. Continuous reassessment during the acute initiation of PE treatment is important to determine the trajectory of disease.

Evaluation of RV structure and function for acute PE

Evidence of RV dilation is important, as it is a marker that the RV is struggling against the elevated pulmonary vascular resistance and afterload. Acute morbidity and mortality from acute PE are largely determined by the evolution of RV failure into cardiogenic/obstructive shock, irrespective of the actual clot burden. Evidence of RV dilation resulted in an odd's ratio (OR) of 4.19 for death from any cause compared to those without RV dilation.

The most common method for evaluating the right ventricle is on the initial CT pulmonary angiogram (CTPA) which diagnosed the acute PE. The following signs on CT scan demonstrate evidence of RV dilation and elevated RV pressure:

Echocardiographic evaluation of the right ventricle can be undertaken with a formal study, or by point-of-care ultrasound (POCUS) by trained practitioners. The following signs on echocardiogram demonstrate evidence of RV dilation, elevated RV pressure, and impaired RV systolic function:

Echocardiogram may additionally reveal the presence of clot-in-transit (CIT) in the right atrium or right ventricle, or the presence of a right-to-left shunt through a patent foramen ovale (PFO).

Risk stratification within the intermediate PE category

Intermediate-risk PE patients have a systolic blood pressure (SBP) over 90 mm Hg and an elevated PESI/sPESI necessitating consideration for admission to hospital. Initial treatment is often with parenteral anticoagulation, usually with low molecular weight (LMWH).

Intermediate-low risk PE has either an elevated troponin level OR evidence of RV dilation, but not both. This subgroup has a better prognosis, with a 30-day mortality around 2% provided they don't worsen while in hospital. A multicenter retrospective stud

the majority of adverse events occurred within 96 hours (4 days) of the even
patients can easily be transitioned to oral anticoagulation within 48 hours if t
shown evidence of progression to intermediate-high risk.

Intermediate-high risk PE patients have both RV dilation and an elevated troponin level, with higher risks of decompensation and death. However, most patients can safely be treated with anticoagulation alone, which remains the standard of care for this cohort of patients. If advanced therapies are considered, anticoagulation should NOT be delayed.

The following clinical features within the intermediate-high risk PE category should raise concerns for risk of clinical deterioration:

- Persistent tachycardia >110 bpm
- Borderline hypotension with SBP 90-100 mmHg
- Shock index (HR/SBP) > 1
- Large central clot burden
- Elevated lactate >2.0 mmol/L
- Hypoxemia or rapidly escalating oxygen requirements (FiO2 >4L NP)
- Presentation with syncope or pre-syncope

In the presence of any of these higher-risk features, in addition to RV dysfunction and elevated troponin, it is reasonable to consider advanced reperfusion therapies in addition to anticoagulation, as discussed below, especially if the patient is not improving (or worsening) with anticoagulation alone.

High-risk PE requires the presence of hemodynamic instability (SBP <90 mm Hg after correction of any hypovolemia) associated with the acute PE diagnosis and is the fundamental determinant of high-risk PE. The evolution of obstructive/cardiogenic shock within the “RV death spiral” frequently results in further hemodynamic deterioration and cardiovascular collapse. **High-risk PE is considered a medical emergency and requires immediate treatment**. Patients with high-risk PE should be treated with a form of reperfusion therapy (systemic thrombolysis, catheter-directed intervention, surgery or extracorporeal membrane oxygenation (ECMO) in addition to anticoagulation.

Treatment of PE:

Whenever possible, it is recommended that a multidisciplinary care team such as a PE response team (PERT) discuss the management of intermediate-high and high-risk PE. PERT teams vary within institutions and can comprise members from ICU, Thrombosis

Consultants, Interventional Radiology, Cardiothoracic Surgery, Respirology, I
Medicine, and others. The benefit of PERT is shared decision making and fa
management for PE patients, including consideration for advanced therapies beyond anticoagulation, provided that is in keeping with the patient's goal of care.

Anticoagulation:

All patients should receive therapeutic dose anticoagulation upon PE diagnosis, regardless of other interventions being considered, unless contraindicated. Those patients who cannot receive anticoagulation will be discussed later.

Classic teaching has been to treat patients with acute PE with unfractionated heparin, as it can be reversed quickly, if needed. However, a retrospective analysis of a PERT reviewed 505 patients and found that *only* 26.3% were in the therapeutic range by 24 hours and *only* 28.4% were therapeutic by 48 hours. Additionally, an observational study of 2610 patients found that UFH was associated with a 5-fold increased risk of moderate to severe bleeding compared to LMWH. A meta-analysis of 1951 patients found that UFH may also be associated with a higher risk of bleeding following thrombolysis.

LMWH is preferred by many centers due to the rapid onset of action. LMWH can be administered once or twice daily (by splitting the daily therapeutic dose into two approximately equal parts, ie, dalteparin 100units/kg bid, enoxaparin 1mg/kg bid, or tinzaparin 175u/kg split into two doses) provided renal function and BMI are appropriate for use.

Intermediate-low risk PE are usually initially treated with LMWH and converted to an oral anticoagulant if they have not deteriorated within 48 hours.

Systemic Thrombolysis (ST)*:*

Systemic thrombolysis (ST) is considered the first-line treatment for patients with **high-risk PE** without any contraindications as endorsed by the 2019 ESC guidelines on PE management. In a 2015 meta-analysis of randomized controlled trials, ST significantly reduced PE-related mortality (OR 0.15). The rate of major bleeding with ST is approximately 9.9%, and the rate of intracranial hemorrhage is approximately 1.7%, both significantly higher than anticoagulation alone.

There are absolute and relative contraindications for ST (**Table 2**) that should be evaluated and considered on an individual basis when deciding on treatment for high-risk or intermediate-high risk PE.

Table 2: Contraindications to ST

Dose regimens of systemic thrombolysis

Multiple ST regimens have been used, with no randomized trials comparing the different ST regimens for effectiveness and safety. The following ST regimens can be considered:

- recombinant tissue plasminogen activator (rt-PA) 100 mg: 10 mg given as a bolus over 5-10 minutes and the remaining 90 mg over 90-140 minutes
- rt-PA 0.6 mg/kg (maximum of 50 mg) over 15 minutes
- rt-PA 0.5 mg/kg to a maximum of 50 mg, given as a 10 mg bolus over 1 minute, then the remainder over 2 hours
- for cardiac arrest: rt-PA 50 mg as a bolus over 1 minute or tenecteplase (weight-based dosing per algorithm) up to 50mg. Refer to UpToDate, product monograph, or hospital formulary.

Intermediate-high risk PE patients are normotensive but are still hemodynamically compromised. Many of these patients demonstrate signs of low cardiac output and hypoperfusion despite normal blood pressure (eg, lactate >2 mmol/L). The rationale for consideration of ST in this subgroup of PE is to resolve RV dysfunction and prevent hemodynamic deterioration. The PEITHO study (2014) randomized 1005 patients with intermediate-high risk PE to receive full-dose tenecteplase or standard anticoagulation alone. Despite a significant reduction in hemodynamic deterioration, there was a 10-fold increase in intracranial hemorrhage, and no difference in overall mortality. The 2019 ESC guidelines recommend AGAINST full-dose ST for patients with intermediate-high risk PE (level IIIB recommendation; should not be considered, moderate quality evidence).

The 2013 MOPETT study used reduced dose (50 mg IV) alteplase in a small group of 121 patients. There was improved mortality without an increased bleeding risk. The ongoing PEITHO-III study is comparing tPA 0.6 mg/kg, up to a maximum of 50 mg IV to anticoagulation alone. Evidence will be emerging on the role of reduced dose ST for intermediate-high risk PE, but the current data does not support routine use at this time; selected use of reduced dose ST in this population can be made on an individual case basis, weighing the risks and benefits. This may be particularly appropriate if further deterioration is strongly anticipated based on high-risk clinical features.

Anticoagulation after systemic thrombolysis

Intravenous heparin infusion: we suggest discontinuing heparin while ST is being administe
thrombolysis, a partial thromboplastin time (PTT) should be drawn 2 hours after ST is complet
be resumed without a bolus if the PTT level is less than 2x the upper limit of normal.

LMWH treatment is not a contraindication to ST. After ST is administered, we suggest either switching to IV heparin for 24 hours or continuing LMWH at next scheduled dose, provided it is more than 2 hours after completion of thrombolysis and there is no evidence for bleeding.

Catheter-based Therapies:

For patients with **high-risk PE**, who have contraindications to ST or failed ST, alternative reperfusion strategies should be considered, including catheter-based therapies (catheter-directed thrombolysis, mechanical thrombectomy) or surgical embolectomy. The 2019 ESC guidelines for this population receive a level IIC recommendation (moderate strength, low quality of evidence). **Intermediate-high risk PE** patients who have deteriorated on anticoagulation may also be considered for advanced therapies (2019 ESC guidelines level IIC recommendation).

Catheter directed thrombolysis (CDT) allows slow and local infusion of thrombolysis drug in low doses (0.5-1 mg/hour/catheter for up to 24 hours) directly to the pulmonary arteries. This can be delivered alone or in combination with mechanical or ultrasound fragmentation of the clot. At the present time, the 2019 ESC guidelines consider CDT a class IIC recommendation, favoring ST first over CDT.

Mechanical thrombectomy (MT) involves percutaneous venous access (commonly femoral) and aspiration of the clot from the pulmonary artery with a suction catheter. This procedure is done by interventional radiology or interventional cardiology depending on local expertise. MT can be feasible in **high-risk PE** patients and can be considered for patients with contraindications to ST or CDT. Due to the large catheter inserted into the vein (usually femoral), anticoagulation should be initiated/continued as clinically appropriate. There should be no delay in anticoagulating a patient for acute PE pending deciding on MT as an added treatment option.

For the **intermediate-high risk PE** patients, observational and small randomized studies have shown a significant improvement in the RV/LV ratio with both CDT and MT. Prospective randomized studies are underway to assess benefit of catheter-based therapies as initial treatment in addition to standard anticoagulation. However, until this evidence is available, anticoagulation is still the treatment of choice. Selected upfront use of catheter-based therapies in this population can be made on an individual case basis, weighing the risks and benefits. This may be particularly appropriate if further deterioration is strongly anticipated based on high-risk clinical features.

Surgical embolectomy is a cardiac surgical procedure where the patient is transitioned to cardiopulmonary bypass for open heart surgery and the PE is removed from the pulmonary arteries. Discussion with cardiothoracic surgeons is always required. Despite successful clot removal, RV dysfunction can persist, which may require the need to prepare for ECMO at the onset of surgery.

ECMO: Veno-arterial (V-A) ECMO can be considered for the patient in hemodynamic collapse following an acute PE. If thrombolysis is contraindicated or fails and the patient is too unstable for interventional procedures, then V-A ECMO should be considered. Cannulation strategy is most commonly achieved through the femoral artery and vein. Recent systemic thrombolysis is NOT a contraindication to ECMO cannulation, but care should be taken within 60 minutes after ST to mitigate bleeding risk. ECMO can improve hemodynamics, restore organ perfusion, improve RV function and stabilize the patient to allow for more definitive treatments. A 2021 DUTCH extracorporeal life support (ECLS) meta-analysis, however, showed no short-term mortality benefit with V-A ECMO although low quality evidence exists for those <60 years of age and those patients having surgical embolectomy.

Special considerations:

Inferior vena cava (IVC) filters

Temporary IVC filters are indicated for patients with DVT and PE who have contraindications t
cases of PE with contraindication to anticoagulation, but without DVT, serial ultrasound may be
avoid IVC filter insertion, resources permitting.

If inserted, IVC filters should be scheduled for removal as soon as anticoagulation is possible. IVC filters can be removed as late as 6-12 months from insertion, but the risk of the filter clotting or becoming embedded in the vessel wall increases with time. Therefore, early IVC filter retrieval within 4 weeks of starting anticoagulation is preferred if no further interruptions in anticoagulation are anticipated.

Pregnancy

LMWH is the treatment of choice for all pregnant persons with PE, including intermediate risk PE. If they are decompensating or have very high-risk features, the multidisciplinary team must include the obstetrical team to plan for treatment options in the event of hemodynamic worsening. Systemic thrombolysis is safe for pregnant patients but carries an increased risk of placental bleeding, peri-partum bleeding, and fetal loss. Consideration of catheter-based therapies can be made for pregnant patients and should involve discussion with obstetrics and anesthesia.

Other Relevant

- Pregnancy: Diagnosis of PE and DVT
- Pregnancy: Thromboprophylaxis
- Pregnancy: Venous Thromboembolism Treatment
- Pulmonary Embolism: Diagnosis
- Pulmonary Embolism: Treatment
- Unfractionated Heparin and Low Molecular Weight Heparin

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